SD-TSIA205

Exercise 1

Bayesian estimation

We assume that for $n \in \{1, \cdots, N\}$ that

$$y(n) = m + w(n)$$

where

- $y(n)$ are the observations,
- $w(n)$ a white zero-mean Gaussian noise with known variance σ^2
- m the **unknown** mean we would like to estimate. We assume that m follows a *a priori* distribution p_M. In addition, m is independent of the process w.

First, we assume that m is zero-mean Gaussian with known variane σ_M^2.

1. Remind the closed-form expression for the MMSE estimate of m denoted by $\hat{m}_N$.
2. Calculate the bias of the estimate.
3. Calcule the MMSE of the estimate.
4. Comment when $\sigma_M^2 \to 0$ or $\to \infty$.

Secondly, we assume that m is now uniformly-distributed between the interval $[-\alpha, \alpha]$ with $\alpha > 0$.

5. Find in closed-form the MMSE estimate.
6. Comment when $\alpha \to 0$ or $\to \infty$.